

DIAGONAL DIFFERENCE:

Problem

Submissions

Leaderboard

Discussions

Given a square matrix, calculate the absolute difference between the sums of its diagonals.

For example, the square matrix *arr* is shown below:

```
1 2 3
4 5 6
9 8 9
```

- The left-to-right diagonal = $1 + 5 + 9 = 15$.
- The right-to-left diagonal = $3 + 5 + 9 = 17$.

Their absolute difference is $|15 - 17| = 2$.

Function description

Complete the *diagonalDifference* function with the following parameter:

- int arr[n][m]*: a 2-D array of integers

Return

- int*: the absolute difference in sums along the diagonals

Input Format

The first line contains a single integer, *n*, the number of rows and columns in the square matrix *arr*.

Each of the next *n* lines describes a row, *arr[i]*, and consists of *n* space-separated integers *arr[i][j]*.

Constraints

- $-100 \leq arr[i][j] \leq 100$

Sample Input

HackerRank

Prepare Algorithms Warmup

Diagonal Difference

```
85 | return s;
86 | }
87 |
88 | vector<string> split(const string &str) {
89 |     vector<string> tokens;
90 |
91 |     string::size_type start = 0;
92 |     string::size_type end = 0;
93 |
94 |     while ((end = str.find(" ", start)) != string::npos) {
95 |         tokens.push_back(str.substr(start, end - start));
96 |     }
```

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Test case 0

Compiler Message

Success

DYNAMIC ARRAY:

HackerRank

PrepareData StructuresArraysDynamic Array

Problem

Submissions

Leaderboard

Discussions

Help

• Declare a 2-dimensional array, `arr`, with `n` empty arrays, all zero-indexed.

• Declare an integer, `lastAnswer`, and initialize it to 0.

You need to process two types of queries:

- Query: `1 x y`
 - Compute `idx = (x ⊕ lastAnswer)`.
 - Append the integer `y` to `arr[idx]`.
- Query: `2 x y`
 - Compute `idx = (x ⊕ lastAnswer)`.
 - Set `lastAnswer = arr[idx][y%size(arr[idx])]`.
 - Store the new value of `lastAnswer` in an answers array.

Notes:

- `⊕` is the bitwise XOR operation, which corresponds to the `^` operator in most languages. Learn more about it on [Wikipedia](#).
- `%` is the modulo operator.
- Finally, `size(arr[idx])` is the number of elements in `arr[idx]`.

Function Description

Complete the `dynamicArray` function with the following parameters:

- `int n`: the number of empty arrays to initialize in `arr`
- `int queries[q][3]`: 2-D array of integers

Returns

- `int[]`: the results of each type 2 query in the order they are presented

Input Format

The first line contains two space-separated integers, `n`, the size of `arr`

```

48     getline(cin, first_multiple_input_temp);
49
50     vector<string> first_multiple_input = split(rtrim(first_multiple_input_temp));
51
52     int n = stoi(first_multiple_input[0]);
53
54     int q = stoi(first_multiple_input[1]);
55
56     vector<vector<int>> queries(q);
57
    
```

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☒ Test case 0
 ☐

☒ Test case 1
 ☐

Compiler Message

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TIME CONVERSION

Problem

Submissions

Leaderboard

Discussions

Full

Given a time in **12-hour AM/PM format**, convert it to military (24-hour) time.

Note: - 12:00:00AM on a 12-hour clock is 00:00:00 on a 24-hour clock.

- 12:00:00PM on a 12-hour clock is 12:00:00 on a 24-hour clock.

Example

- `s = '12:01:00PM'`
Return '12:01:00'.
- `s = '12:01:00AM'`
Return '00:01:00'.

Function Description

Complete the `timeConversion` function with the following parameter(s):

- `string s`: a time in 12 hour format

Returns

- `string`: the time in 24 hour format

Input Format

A single string `s` that represents a time in 12-hour clock format (i.e.: hh:mm:ssAM or hh:mm:ssPM).

Constraints

- All input times are valid

Sample Input 0

07:05:45PM

22

23

24

25

26

27

28

29

30

31

32

```
1 if (hour == 12) {
2     hour += 12;
3     s[0] = '0' + (hour / 10);
4     s[1] = '0' + (hour % 10);
5 }
6
7 s.erase(8, 2);
8
9 return s;
10 }
```

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Problem Solving

15.00 points!

You are now 60 points away from the 2nd star for your problem solving badge.

14%

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Test case 0

Compiler Message

Test case 1

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COMPARE THE TRIPLETS:

Problem

Submissions

Leaderboard

Discussions

Help

HackerRank | Prepare Algorithms Warmup Compare the Triplets

- Priority: for elements i , $a[i] > b[i]$ so Alice receives a point.

The return array is `[1, 1]` with Alice's score first and Bob's second.

Function Description

Complete the function `compareTriplets` with the following parameter(s):

- `int a[3]`: Alice's challenge rating
- `int b[3]`: Bob's challenge rating

Returns

- `int[2]`: the first element is Alice's score and the second is Bob's score

Input Format

The first line contains 3 space-separated integers, $a[0]$, $a[1]$, and $a[2]$, the respective values in triplet `a`.

The second line contains 3 space-separated integers, $b[0]$, $b[1]$, and $b[2]$, the respective values in triplet `b`.

Constraints

- $1 \leq a[i] \leq 100$
- $1 \leq b[i] \leq 100$

Sample Input 0

```
5 6 7
3 6 10
```

Sample Output 0

```
1 1
```

```
27 |         }
28 |     }
29 | }
30 |
31 | return {alice, bob};
32 | }
33 |
34 | int main()
35 | {
36 |     ofstream fout(getenv("OUTPUT_PATH"));
37 |
38 |     string a_temp_temp;
```

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29%

Congratulations

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Test case 0

Compiler Message

Test case 1

Success

Sparse Array:

```
36 {
37     ofstream fout(getenv("OUTPUT_PATH"));
38
39     string stringList_count_temp;
40     getline(cin, stringList_count_temp);
41
42     int stringList_count = stoi(ltrim(rtrim(stringList_count_temp)));
43
44     vector<string> stringList(stringList_count);
45
46     for (int i = 0; i < stringList_count; i++) {
```

Line: 33 Col: 2

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64%

75/100

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Next Challenge

Test case 0 

Test case 1 

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